Total number of printed pages–3

1 SEM MCA (NCBCS) 3BE

2015

(December)

COMPUTER APPLICATION

Paper : MCA–103

(Basic Electronics)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer question **No. 1** and **2** and **any four** from the rest.*

1. (a) Convert $(0{\cdot}1010011)_2$ into decimal. 1

(b) Convert $(101011011)_2$ into octal. 1

(c) Convert $(105B)_{16}$ into binary. 1

(d) What is the value of 8 in 8–4–2–1 code? 1

(e) Subtract (111–101) using 1's complement method. 1

Contd.

111
010
1001
016

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2. (a) Design a logic circuit to get

$X = (A\overline{B}+\overline{A}C\ (\overline{AC+C})$ 5

(b) What are universal gates and why they are said so? 3

(c) Using truth table show that the boolean expression $AB+A\overline{B} = A$ 2

3. (a) Simplify the following boolean expression 5

$$\overline{\overline{AB}+\overline{A\overline{C}+\overline{B\overline{C}}}+\overline{C}}$$

(b) Why a gray code called unit distance code? Design a BCD to gray code converter. 1+7=8

(c) Explain the boolean function $F = x+yz$ as a sum of minterms. 2

4. (a) What is multiplexer? Realize a 8×L multiplexer. 8

(b) Using Kaunaugh map reduce the following boolean function and draw the logic diagram using only AND-OR gate. 7

$F(w, x, y, z) = \pi(0, 1, 3, 6, 7, 9, 11, 12)$

5. (a) Design a full subtractor using 3×8 decoder. 7

(b) Draw and explain the working principle of T-flip-flop. 5

(c) What is full adder? 3

6. (a) What is decoder? Realise a 4×10 decoder. 10

(b) Explain half adder circuit with diagram. 5

7. Write short notes on: ***(any three)*** 3×5=15

(a) D-Flip-flop

(b) Fan-in and fan-out capacity

(c) CPU organisation

(d) Primary and Secondary memory.

(e) Combination *vs* sequential circuit.

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Total No. of printed pages = 4

1 MCA 3 BEL

2012

(December)

BASIC ELECTRONICS

Paper : 103

Full Marks – 75

Pass Marks – 34

Time – Three hours

The figures in the margin indicate full marks for the questions.

Attempt any *five* questions.

1. (a) Convert $(101{\cdot}25)_{10}$ into binary number. 3

(b) Give the electrical realization of AND gate. 3

(c) Prove that $A+\overline{A}B = A+B$. 3

(d) What are universal gates ? Why they are said so ? 1+2=3

[Turn over

A.1+Ā.B

R)

B)

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(c) Realize exclusive OR gate with basic gate. Why this gate is said as inequality detector ? 2+1=3

2. (a) Minimize the following expression : 5

$$Y = \left(\overline{\overline{A}B}\right) + \left(\overline{A\overline{C} + \overline{B}\overline{C}}\right) + C$$

(b) Write the following expression in canonical SOP form : 5

$$Y = f(A,B,C) = A\overline{B} + \overline{B}C + \overline{A}\overline{C}$$

(c) What is BCD code ? Explain. 5

3. (a) Realize the following expression : 4+4=8

$$Y = A\overline{B} + \overline{A}\overline{C}$$

with (i) NAND Gate

(ii) NOR Gate

(b) Minimize using Karnaugh map : 7

$$f(A,B,C,D) = \sum(0,2,8) + \sum d(4,6,10,13)$$

4. (a) Differentiate between combinational circuit and sequential circuit. 4

(b) Design and explain a (3×8) line decoder circuit. 7

(c) What is a demultiplexer ? How it is different from a decoder circuit ? 3+1=4

5. (a) What is multiplexer ? Realise a 4×1 multiplexer. 2+6=8

(b) Explain S-R flip-flop. What is the disadvantage in S-R flip-flop ? How this can be overcome ? 4+2+1=7

6. (a) Design a synchronous counter with T-flip-flop for the following sequences : 10

$$6 \rightarrow 2 \rightarrow 0 \rightarrow 5 \rightarrow 4$$

(b) Explain D-flip-flop with its state table and excitation table. 5

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7. (a) Define the following terms : 2×3=6

(i) Fan in capacity

(ii) Fan out capacity

(iii) Figure of merit

(b) What are the advantages of C-MOS logic family ? 4

(c) What are registers ? Explain SIPO type of register. 5

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1 SEM MCA (NCBCS) BE 3

2016

(December)

COMPUTER APPLICATION

(Basic Electronics)

Paper : 103

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

Answer Q. ***1*** *and* ***any four*** *from the rest.*

1. *(a)* Convert $(2B6D)_{16}$ into decimal. 2

   (b) Convert $(443)_{10}$ into octal. 1

   (c) Convert $(1011{\cdot}1011)_2$ to its equivalent octal number. 2

   (d) Subtract $(1001)_2$ from $(1011)_2$ by 2's complement method. 2

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(e) How many control signal is needed for a 16 × 1 multiplexer? 1

(f) What do you mean by Random access and serial access? 3

(g) Why is the main memory called volatile in nature? 1

(h) What do you mean by synchronous and asynchronous data transfer? 3

2. (a) Briefly explain the gates for two or three variables with truth table and logic diagram. 7

(b) Explain : 2×4=8

(a) Parity code

(b) Hamming code and distance

(c) ASCII

(d) Cycle redundancy code.

3. (a) What is *k*-map and don't care conditions? Using *k*-map reduce the following Boolean functions and draw the logic diagram

$$F(A, B, C) = \sum (3, 4, 6, 7).$$ 7

(b) Implement $Y = \overline{AB} + A + \overline{(B + C)}$ using NAND gates only. 4

(c) What are the characteristic of Diode and Transistor? 4

4. (a) Briefly explain semiconductor memory and auxiliary memory. 7

(b) Describe *three* state buffer with a decoder. 6

(c) What are the differences between half adder and full adder? 2

5. (a) What do you mean by Asynchronous and Synchronous counter?

Briefly describe Ripple counter. 6

(b) What is the difference between R-S and J-K flip flop? Explain J-K flip flop. 6

(c) Explain how demultiplexer works? 3

6. (a) How decoder is differ from encoder? Briefly describe 3 to 8 decoder. 6

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(b) What do you mean by register load? What is the purpose of shift register? 3

(c) Briefly describe multiplication algorithm with a suitable example. 6

7. (a) Briefly explain functionalities of different blocks of CPU. 7

(b) Explain the concept of TTL. 3

(c) Explain working principle of adder — subtractor. 5

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Total number of printed pages–4

1 SEM MCA (NCBCS) 3

2018

(December)

COMPUTER APPLICATION

Paper : 103

(Basic Electronics)

Full Marks : 75

Pass Marks : 34

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **any five** questions.*

(a) What is the binary equivalent of the $(2B7A)_{16}$? 2

(b) $(11010{\cdot}11010)_2 = (?)_{10}$ 2

(c) Subtract $(110)_2$ from $(100)_2$ by 2's complement subtraction method. 2

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(d) A 7-bit hamming code is received as 0101101. What is the correct code? 2

(e) State the principle of duality. Prove the following: $AB + AB' = A$. 2

(f) Draw the following logic diagram with NAND gates only: 5

$F = (A+C)(B'+D')(A'+B'+C')$

2. (a) Obtain the canonical SOP form of the following: 2

$F(A, B, C, D) = AB + ACB$

(b) What do you mean by don't care conditions? Using K-map, reduce the following: 2+3=5

$F(W,X,Y,Z) = \sum(0,1,3,6,7,9,11,12)$

(c) Simplify the following and draw the logic diagram: 5

$X = (AB' + A'C)(AC + C)'$

(d) State the differences between decoder and encoder. 3

3. (a) Describe about Master-Slave J-K flip-flop. 5

(b) Design a full-adder circuit. 5

(c) Describe the A/D converter and what are its characteristics? 2+3=5

4. (a) Describe the function of an 8-to-1 line multiplexer with its proper diagram. 5

(b) Explain the working of a T flip-flop. 5

(c) What are the differences between combinational and sequential circuit? 5

5. (a) What is a decoder? Realise a 3-to-8 line decoder with its logic diagram. 5

(b) Explain the working of an asynchronous counter with its timing diagram. 5

(c) What are the characteristics of digital ICs? 5

6. (a) Explain the working of shift-right SISO registers. 5

(b) Using K-map and don't care condition simplify the following: 5

$F(A, B, C, D) = \sum(0, 2, 3, 6, 7) + d(8, 10, 11, 15)$

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(c) What are the differences between *BCD* and excess-3 code? Convert the $(10101101)_2$ into its gray code. 2+3=5

7. Write a short note on the following: 5×3=15
(any three)

(a) Different types of Registers

(b) S-R flip-flop

(c) Full subtractor

(d) Demultiplexer

(e) Minterms and Maxterms.

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1 SEM MCA (NCBCS) 103

2019

(December)

COMPUTER APPLICATION

Paper : MCA-103

(Basic Electronics)

Full Marks : 75

Time : Three hours

The figures in the margin indicate full marks for the questions.

Answer Question No. ***1*** *and* ***any four*** *from the rest.*

1. *(a)* Convert $(362.66)_8$ into decimal. 2

(b) Convert $(1064)_{10}$ into hexadecimal. 2

(c) Convert $(10111)_2$ into decimal. 2

(d) Do the following : 2
1111+101+11 = ?

Contd.

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(e) $(1110)_2 - (1100)_2$ using 2's complement subtraction method. 2

(f) Convert $(367)_{10}$ into its excess-3 code. 2

(g) What is the principle of duality? Write de-Morgan's theorem. 3

2. (a) Prove that, $A+A=A$. 2

(b) Define Minterm and Maxterm with proper examples. 4

(c) Obtain the canonical SOP form of the following : 3

$$F(A, B, C, D) = AB + ACD$$

(d) Realise the function using NAND gates only : 4

$$F = A + BCD'$$

(e) Find the complement of the following expression : 2

$$F = A(B + C)(C' + D')$$

3. (a) Simplify the expression using Karnaugh map and draw the expression : 6

$$F(A, B, C, D) = \Sigma(7, 9, 10, 11, 12, 13, 14, 15)$$

(b) Design a 3-to-8 line decoder using 2-to-4 line decoder. 5

(c) Differentiate between adders and subtractors. 4

4. (a) What is multiplexer? Describe a 4-to-1 line multiplexer with proper logic diagram. 6

(b) Write the functionality of a master-slave flip-flop. 5

(c) Differentiate between combinational and sequential circuits. 4

5. (a) Describe SISO registers in detail. 5

(b) Differentiate between J-K and T flip-flops. 5

(c) Construct a full adder using 2 half adders. 5

6. (a) Using K-map simplify the following : 5

$$F(A, B, C, D) = \Sigma(1, 5, 7, 10, 11) + d(2, 3, 6, 13)$$

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(b) Differentiate between asynchronous and synchronous counters. 5

(c) What do you mean by Integrated Circuits (IC)? Discuss about various categories of ICs. 5

7. Write short notes on : ***(any three)*** 5×3=15

(a) Encoder

(b) Hamming codes

(c) Types of registers

(d) Working of demultiplexer

(e) Binary to Gray code converter.

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